

Chapter 9: The number plane

6 lessons · one lesson for each exercise · each lesson is
one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-LIN-C-01 creates and
displays number patterns and graphs linear relationships on the
Cartesian plane.

CHAPTER

9

9.01 Grid references on maps

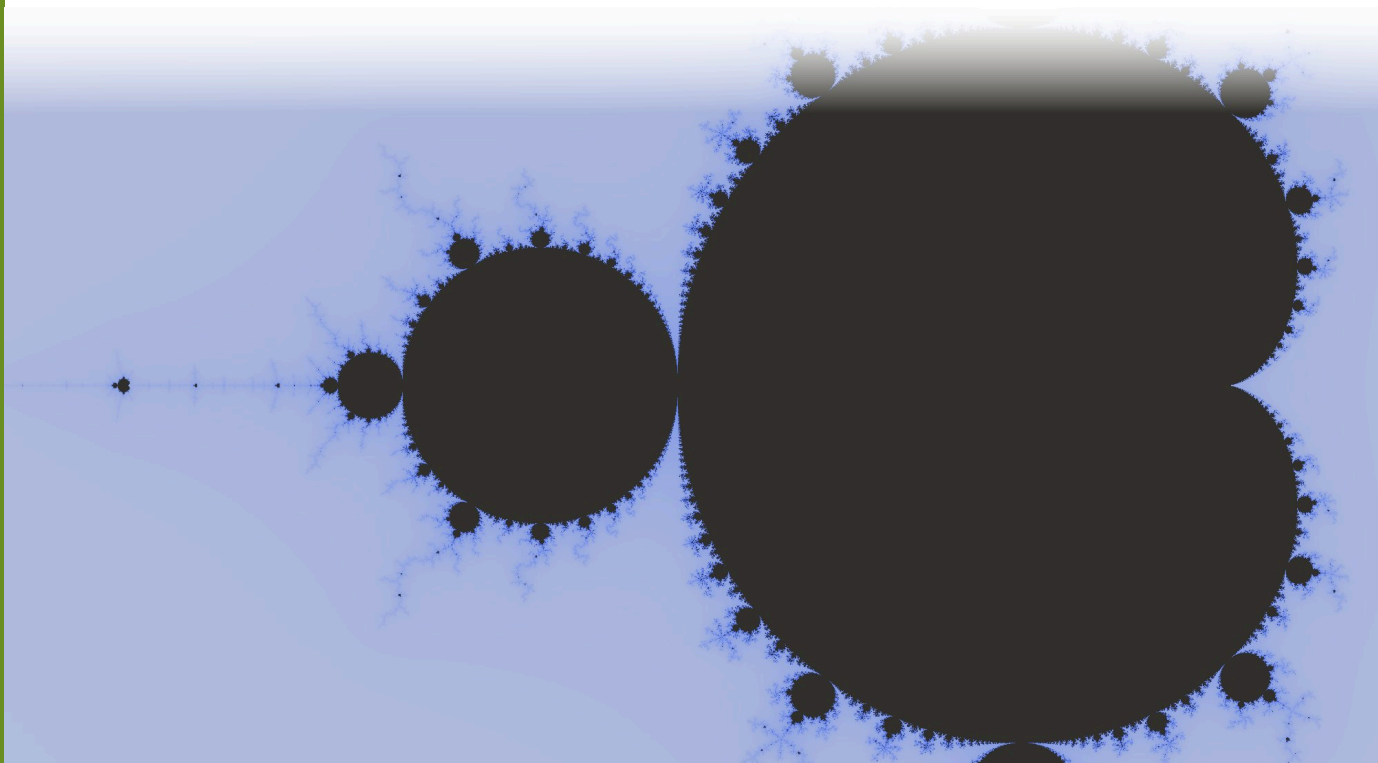
9.02 Points in the first quadrant

9.03 The number plane

9.04 Plotting tables of values

9.05 Tables of values from rules

9.06 Graphing linear equations



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

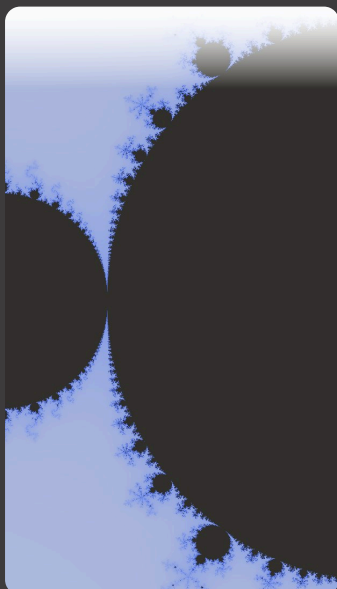
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
9.01	Grid references on maps	3		☐
9.02	Points in the first quadrant	11		☐
9.03	The number plane	19		☐
9.04	Plotting tables of values	27		☐
9.05	Tables of values from rules	35		☐
9.06	Graphing linear equations	43		☐

In this chapter you will

- use grid references on a map
- plot and read points in the first quadrant
- plot and read points in all four quadrants
- plot points from tables and describe patterns
- complete tables of values from rules
- graph linear equations and test whether a point is on a line



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the whole set, drawn on the complex number plane. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
9.01

Grid references on maps

Date _____

Learning intentions

We are learning to:

- use grid references to find and describe locations on a map

Success criteria

I can:

- ☐ read a grid reference: letter (across) then number (up)
- ☐ state the feature at a grid reference
- ☐ write the grid reference of a feature

Key terms Write the definition as your teacher explains it.

Grid reference

Map

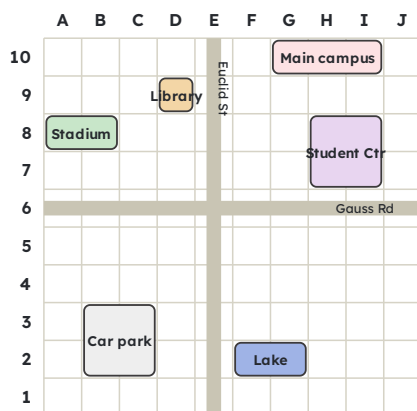
Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the map. What is at each grid reference?



a What is at F2?

Your teacher works this out. Copy every step.

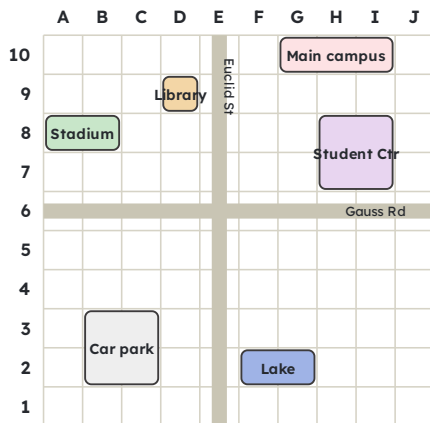
b What is at D9?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the map. What is at:

4 QUESTIONS

REMEMBER A grid reference names a square.



1 F2

2 D9

3 A8

4 H7

YOU DO On your own: **Easy** Set A2. Quick drill: use the map in Set A. What is at:

6 QUESTIONS

1 B2

2 C3

3 G2

4 B8

5 I8

6 H10

YOU DO On your own: **Easy** Set B. Write a grid reference for:

6 QUESTIONS

1 the library

4 the car park

2 the lake

5 the student centre

3 the stadium

6 the main campus

SKILL DRILL 1 Easy basics: practise until it is automatic

24 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: use the map. What is at each grid reference?

6	Wharf	Fire		Post		Vet		Hotel
5					Club			Beach
4	Farm		Mall				Oval	
3				Gym			Cafe	
2		Shop			Bus			Zoo
1	Pool		Bank			Park		
	A	B	C	D	E	F	G	H

1 G3

2 B6

3 F1

4 A6

5 D3

6 D6

7 H2

8 A1

9 A4

10 C1

11 E2

12 G4

ROUND 2 Questions 13–24: write the grid reference of each place.

6	Wharf	Fire		Post		Vet		Hotel
5					Club			Beach
4	Farm		Mall				Oval	
3				Gym			Cafe	
2		Shop			Bus			Zoo
1	Pool		Bank			Park		
	A	B	C	D	E	F	G	H

13 Club

14 Cafe

15 Bus

16 Park

17 Pool

18 Zoo

19 Shop

20 Vet

21 Beach

22 Hotel

23 Fire

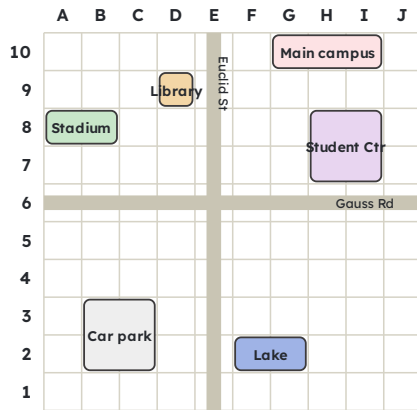
24 Mall

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the map. Write the grid references.



a Write the grid reference of the library.

Your teacher works this out. Copy every step.

b Which grid squares does the car park cover?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use the map.

6 QUESTIONS

REMEMBER Letter first (across), then number (up).

1 Name a grid square that Gauss Road passes through.

3 Where do Gauss Road and Euclid Street meet?

5 How many squares is the lake from the car park, across?

2 Name a grid square that Euclid Street passes through.

4 Which feature is furthest north (highest row)?

6 Which features are in column H?

SKILL DRILL 2 Medium basics: keep practising

20 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 20 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–10: start at the place given and move. Where do you end up?

6	Wharf	Fire		Post		Vet		Hotel
5				Club				Beach
4	Farm		Mall				Oval	
3				Gym			Cafe	
2		Shop			Bus			Zoo
1	Pool		Bank			Park		
	A	B	C	D	E	F	G	H

1 Farm: 6 right, 2 up

2 Gym: 3 left, 2 up

3 Zoo: 4 left, 2 up

4 Pool: 5 right, 2 up

5 Bank: 1 right, 3 up

6 Post: 4 right, 5 down

7 Bus: 2 left, 3 up

8 Park: 2 right, 5 up

9 Wharf: 1 right, 4 down

10 Oval: 2 left, 1 up

ROUND 2

Questions 11–20: how many squares right (or left) and up (or down) from the first place to the second?

6	Wharf	Fire		Post		Vet		Hotel
5				Club				Beach
4	Farm		Mall				Oval	
3				Gym			Cafe	
2		Shop			Bus			Zoo
1	Pool		Bank			Park		
	A	B	C	D	E	F	G	H

11 Fire to Post

12 Club to Bus

13 Mall to Hotel

14 Bus to Gym

15 Beach to Fire

16 Hotel to Zoo

17 Shop to Cafe

18 Pool to Bank

19 Cafe to Park

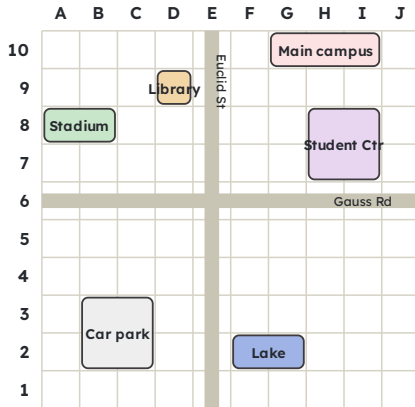
20 Vet to Pool

Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Use the map, then explain.



a Describe how to walk from the lake to the library along the roads.

Your teacher works this out. Copy every step.

b Why is the letter written before the number?

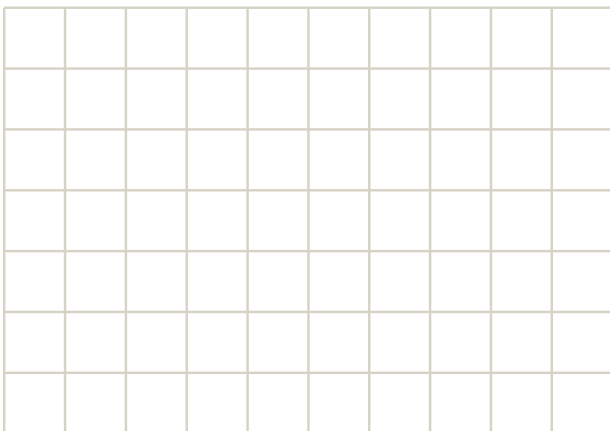
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Draw your own.

2 QUESTIONS

REMEMBER Large features may cover several squares.

1 Draw a map of your school on the grid. Mark 5 places and list their grid references.



2 Write directions from one place to another using grid references.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

A treasure is 3 squares right and 4 squares up from B2. Where is it?

1. 3 squares right of B is E
2. 4 squares up from 2 is 6

Answer: E6

YOU DO

On your own: read the worked example, then try these.

- E1** From H7, go 5 left and 2 down. Where are you?

- E2** How many squares is it from A1 to J10 if you can only move up or right?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is at A8?

Level 2

Write the grid reference of the lake.

Level 3

From D9 go 4 right and 2 down. What is there?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A grid reference names a square.
 2. Letter first (across), then number (up).
 3. Large features may cover several squares.
 4. Check by running your finger across, then up.

Level 1

What is at F2?

1. Column F, row 2

Answer: the lake

Level 2

Library

1. Column D, row 9

Answer: D9

Level 3

Car park

1. Covers 4 squares

Answer: B2, C2, B3, C3

Extension

3 right, 4 up from B2

1. B → E, 2 → 6

Answer: E6

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.02

Points in the first quadrant

Date _____

Learning intentions

We are learning to:

- plot and read points on a number plane in the first quadrant

Success criteria

I can:

- ☐ know the **x-coordinate** is across and the **y-coordinate** is up
- ☐ plot a point (x, y)
- ☐ read the coordinates of a point

Key terms Write the definition as your teacher explains it.

Coordinates

Origin

Axis

Notes

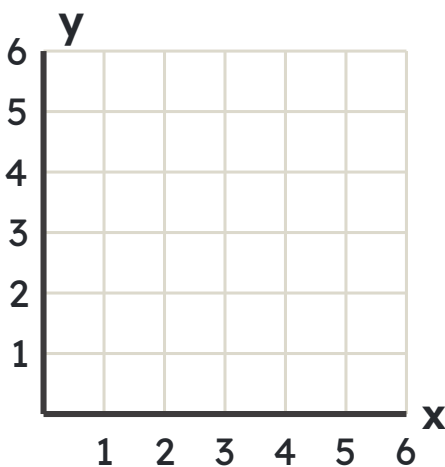
Level 1 · Getting started

Easy · 14 questions

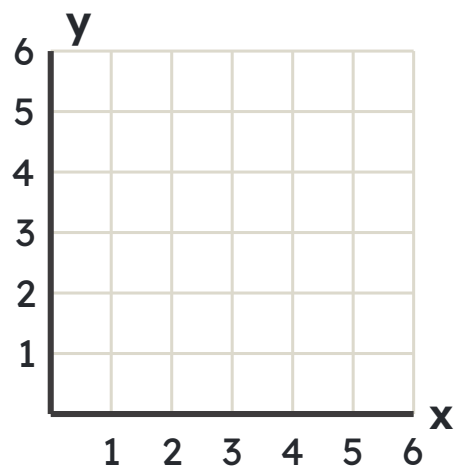
WE DO With your teacher: copy what they write on the board.

Example 1 Plot and label the points.

a Plot A(2, 3) and B(1, 6).



b Plot C(3, 0) and D(0, 4).

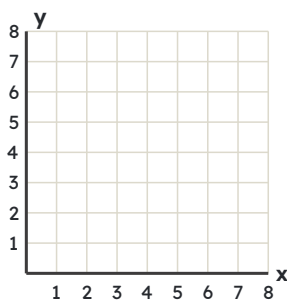


YOU DO On your own: **Easy** Set A. Plot and label each point on the same number plane.

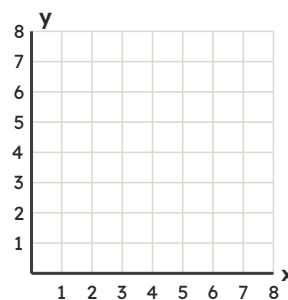
2 QUESTIONS

REMEMBER (x, y): x across first, then y up.

1 A(2, 3), B(1, 6), C(3, 0)



2 D(0, 4), E(5, 2), F(6, 5)



YOU DO On your own: **Easy** Set A2. Quick drill: write the x-coordinate (x) or the y-coordinate (y).

6 QUESTIONS

1 x of (4, 7)

2 y of (4, 7)

3 x of (0, 3)

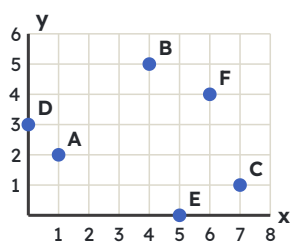
4 y of (6, 0)

5 y of (2, 9)

6 x of (8, 1)

YOU DO On your own: **Easy** Set B. Write the coordinates of each point.

6 QUESTIONS



1 A

2 B

3 C

4 D

5 E

6 F

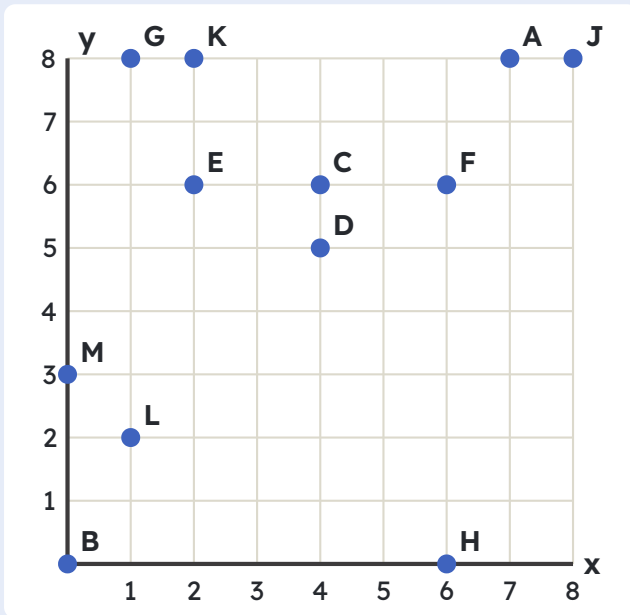
SKILL DRILL 1 Easy basics: practise until it is automatic

28 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: write the coordinates of each point.



1 A

2 B

3 C

4 D

5 E

6 F

7 G

8 H

9 J

10 K

11 L

12 M

ROUND 2 Questions 13–28: write the x-coordinate (x) or the y-coordinate (y) of the point.

13 y of (2, 6)

14 y of (2, 10)

15 y of (12, 4)

16 x of (1, 9)

17 y of (7, 12)

18 x of (12, 0)

19 x of (6, 4)

20 x of (7, 9)

21 x of (8, 0)

22 x of (4, 1)

23 y of (9, 10)

24 x of (0, 10)

25 x of (0, 7)

26 y of (0, 3)

27 y of (3, 7)

28 x of (10, 11)

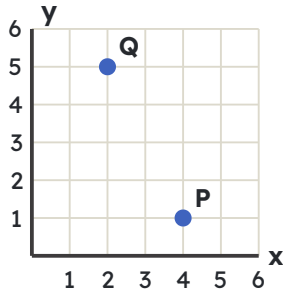
Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Read the coordinates, then explain.

a Write the coordinates of P and Q.



b Is (2, 5) the same point as (5, 2)? Explain.

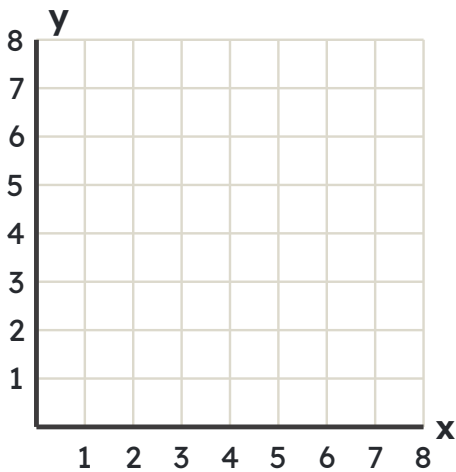
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Plot, join in order and name the shape.

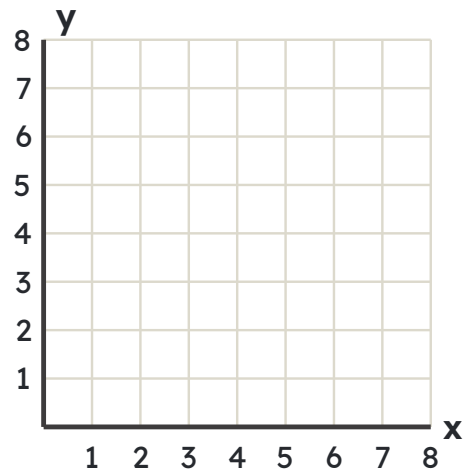
2 QUESTIONS

REMEMBER Start at the origin (0, 0).

1 (1, 1), (6, 1), (6, 4), (1, 4), (1, 1)



2 (2, 1), (6, 1), (4, 6), (2, 1)



SKILL DRILL 2 Medium basics: keep practising

24 questions

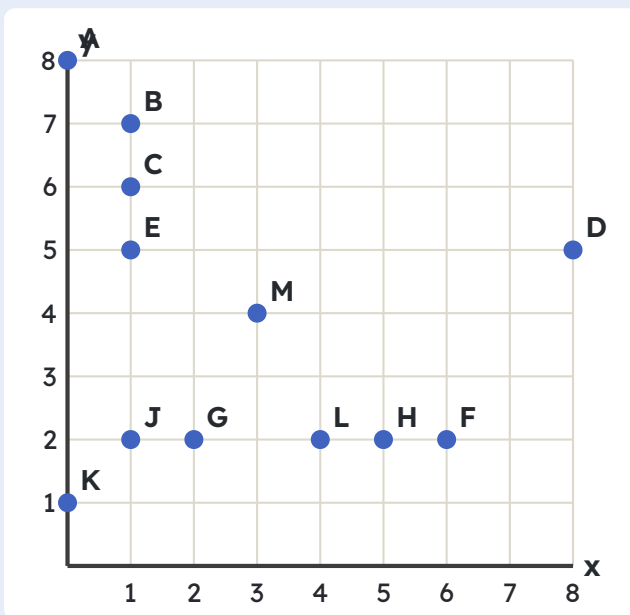
YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: which point is at these coordinates?



1 (1, 7)

2 (1, 2)

3 (3, 4)

4 (6, 2)

5 (1, 6)

6 (1, 5)

7 (8, 5)

8 (5, 2)

9 (0, 1)

10 (4, 2)

11 (0, 8)

12 (2, 2)

ROUND 2

Questions 13–24: the two points are on the same horizontal or vertical line. How far apart are they?

13 (0, 6) and (0, 2)

14 (3, 7) and (6, 7)

15 (1, 9) and (1, 1)

16 (4, 10) and (7, 10)

17 (0, 7) and (5, 7)

18 (2, 8) and (4, 8)

19 (8, 6) and (8, 5)

20 (5, 6) and (5, 3)

21 (9, 5) and (7, 5)

22 (1, 3) and (1, 5)

23 (2, 0) and (2, 3)

24 (8, 7) and (5, 7)

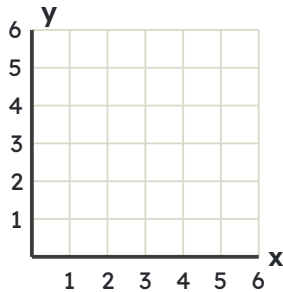
Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Plot, join and name the shape.

- a** Plot $(1, 1)$, $(5, 1)$, $(5, 4)$, $(1, 4)$ and join them. Name the shape.



- b** What are the coordinates of the origin?

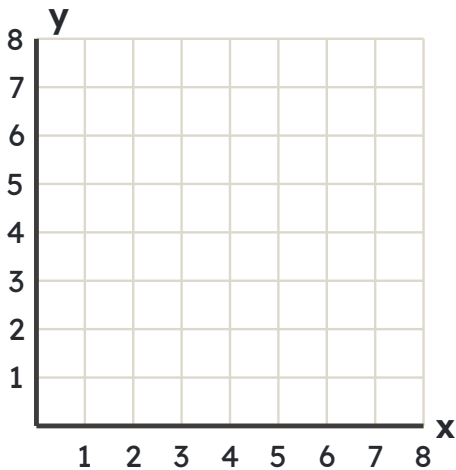
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

2 QUESTIONS

REMEMBER Points on the x-axis have $y = 0$.

- 1** Three corners of a square are $(1, 2)$, $(5, 2)$ and $(5, 6)$. Plot them and find the fourth corner.



- 2** Name 3 points on the x-axis and 3 on the y-axis. What do you notice?



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

Find the midpoint of (2, 3) and (6, 7).

1. Halfway across: $(2 + 6) \div 2 = 4$
2. Halfway up: $(3 + 7) \div 2 = 5$

Answer: (4, 5)

YOU DO

On your own: read the worked example, then try these.

- E1 Find the midpoint of (1, 8) and (5, 2).

- E2 A rectangle has corners (1, 1) and (7, 5). Find its area in square units.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is the x-coordinate of (7, 3)?

Level 2

Where is the point (0, 5)?

Level 3

A rectangle has corners (1, 2), (6, 2), (6, 5). Find the fourth corner.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. (x, y) : x across first, then y up.
 3. Points on the x -axis have $y = 0$.

2. Start at the origin $(0, 0)$.
4. Points on the y -axis have $x = 0$.

Level 1

Plot $A(2, 3)$

1. 2 across, 3 up

Answer: a dot labelled A

Level 2

Coordinates of P

1. 4 across, 1 up

Answer: $(4, 1)$

Level 3

$(2, 5)$ vs $(5, 2)$

1. Order matters

Answer: different points

Extension

Midpoint of $(2, 3)$, $(6, 7)$

1. Average x and y

Answer: $(4, 5)$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.03

The number plane

Date _____

Learning intentions

We are learning to:

- plot and read points in all four quadrants

Success criteria

I can:

- ☐ name the four quadrants
- ☐ plot points with negative coordinates
- ☐ state which quadrant a point is in from its signs

Key terms Write the definition as your teacher explains it.

Quadrant

Origin

Notes

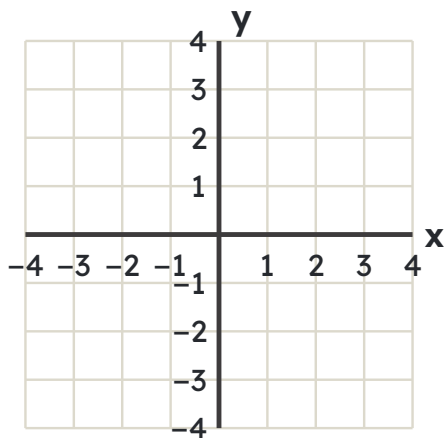
Level 1 · Getting started

Easy · 21 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Label the quadrants, then name the quadrant.

a Label the quadrants 1st to 4th.



b In which quadrant is $(-3, -3)$?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: Which quadrant or axis is each point on?

15 QUESTIONS

REMEMBER The axes cross at the origin (0, 0).

1 (–3, –3)

2 (4, 2)

3 (–1, 5)

4 (3, –6)

5 (0, 4)

6 (–2, 0)

Questions 7–15: which quadrant or axis?

7 (2, –5)

8 (–4, –1)

9 (–3, 6)

10 (0, –2)

11 (5, 5)

12 (–7, 0)

13 (1, –1)

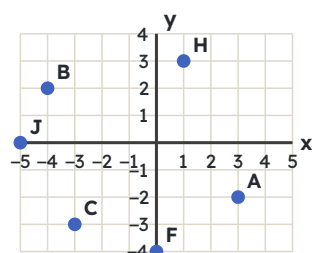
14 (–2, 3)

15 (0, 0)

YOU DO

On your own: **Easy** Set B. Write the coordinates of each point.

6 QUESTIONS



1 H

2 B

3 C

4 A

5 F

6 J

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: which quadrant (1st, 2nd, 3rd or 4th) or which axis is the point on?

1 $(-2, -9)$

2 $(-2, 8)$

3 $(-6, 9)$

4 $(4, -8)$

5 $(6, -5)$

6 $(-6, -4)$

7 $(-7, -6)$

8 $(-1, 7)$

9 $(-8, 4)$

10 $(1, -5)$

11 $(-4, 7)$

12 $(-2, -8)$

13 $(5, 4)$

14 $(-5, 1)$

15 $(2, 3)$

16 $(9, -7)$

17 $(-1, 1)$

18 $(7, -9)$

19 $(4, 8)$

20 $(-4, -6)$

21 $(-7, 1)$

22 $(-5, -8)$

23 $(-2, 3)$

24 $(-6, 0)$

ROUND 2 Questions 25–36: write the coordinates of each point.

25 A

26 B

27 C

28 D

29 E

30 F

31 G

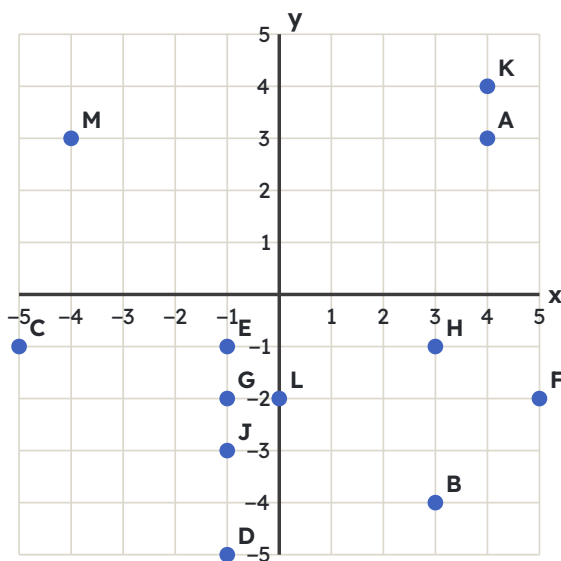
32 H

33 J

34 K

35 L

36 M



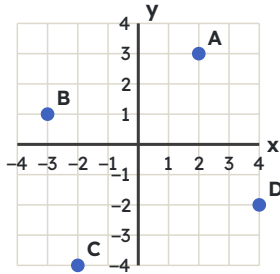
Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Read the coordinates, then name the quadrant.

a Write the coordinates of each point.



b Which quadrant has a positive x-coordinate and a negative y-coordinate?

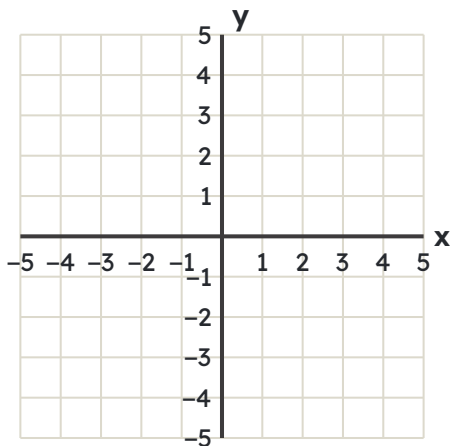
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Plot and join.

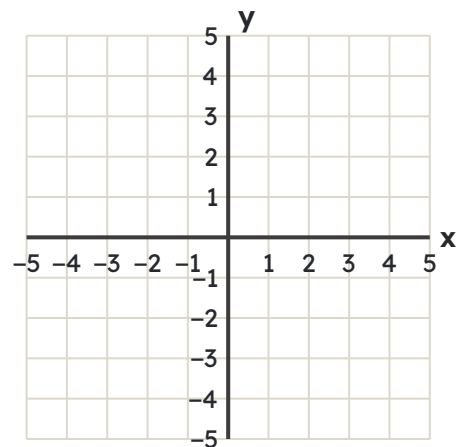
2 QUESTIONS

REMEMBER 1st (+, +), 2nd (-, +), 3rd (-, -), 4th (+, -).

1 (1, 3), (2, 3), (2, 2), (3, 1), (2, 0), (2, -1), (1, -1), (0, -2), (-1, -1), (-2, -1), (-2, 0), (-3, 1), (-2, 2), (-2, 3), (-1, 3), (0, 4), (1, 3). What shape is it?



2 Plot A(-4, -2), B(2, -2), C(2, 3). Find D so ABCD is a rectangle.



SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: reflect the point in the axis given.

- 1 $(-7, -7)$ in the y-axis 2 $(-3, 3)$ in the y-axis 3 $(4, 2)$ in the y-axis 4 $(-4, -1)$ in the x-axis

- 5 $(1, 4)$ in the y-axis 6 $(4, -6)$ in the x-axis 7 $(7, -3)$ in the y-axis 8 $(4, 4)$ in the y-axis

- 9 $(-4, -8)$ in the y-axis 10 $(-2, -6)$ in the y-axis 11 $(8, -7)$ in the x-axis 12 $(5, -6)$ in the x-axis

- 13 $(6, 8)$ in the x-axis 14 $(-2, 4)$ in the x-axis 15 $(-1, -8)$ in the x-axis 16 $(5, 2)$ in the y-axis

ROUND 2 Questions 17–28: which quadrant is described?

- 17 both positive 18 $x = 0$ 19 right of the y-axis, above the x-axis

- 20 left of the y-axis, above the x-axis 21 right of the y-axis, below the x-axis 22 x positive, y negative

- 23 x negative, y positive 24 $y = 0$ 25 both negative

- 26 $(-a, a)$ for a positive number a 27 left of the y-axis, below the x-axis 28 x and y are both zero

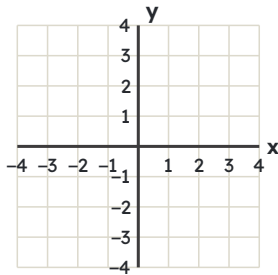
Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Plot and join, then write points.

- a** Plot $(1, 3)$, $(2, 2)$, $(3, 1)$, $(2, 0)$, $(1, -1)$, $(0, -2)$, $(-1, -1)$ and join.



- b** Write a point in the 2nd quadrant and a point on the y-axis.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Quadrants go anticlockwise from top right.

- 1** In which quadrant is a point with a negative x-coordinate and a positive y-coordinate? A 1st B 2nd C 3rd D 4th

- 3** Reflect $(3, -2)$ in the y-axis. Write the new point.

- 2** What are the coordinates of the origin?

- 4** What is the distance between $(-3, 2)$ and $(4, 2)$?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

A square has corners $(-2, -1)$ and $(3, -1)$ along its bottom. Find the other two corners (above).

1. Side length = $3 - (-2) = 5$
2. Go up 5: $y = -1 + 5 = 4$

Answer: $(3, 4)$ and $(-2, 4)$

YOU DO

On your own: read the worked example, then try these.

- E1 A triangle has corners $(-3, 0)$, $(3, 0)$ and $(0, 4)$. Find its area.

- E2 Reflect the triangle $(1, 1)$, $(4, 1)$, $(1, 3)$ in the x-axis. Write the new corners.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

In which quadrant is $(-4, 2)$?

Level 2

Write the coordinates of a point in the 4th quadrant.

Level 3

Reflect $(-2, 5)$ in the x-axis.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. The axes cross at the origin (0, 0).
 2. 1st (+, +), 2nd (–, +), 3rd (–, –), 4th (+, –).
 3. Quadrants go anticlockwise from top right.
 4. Points on an axis are in no quadrant.

Level 1

(–3, –3)

1. Both negative

Answer: 3rd quadrant

Level 2

Point B

1. 3 left, 1 up

Answer: (–3, 1)

Level 3

+x, –y

1. Bottom right

Answer: 4th quadrant

Extension

Square on (–2, –1), (3, –1)

1. Side 5, go up

Answer: (3, 4), (–2, 4)

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.04

Plotting tables of values

Date _____

Learning intentions

We are learning to:

- plot points from a table and describe the pattern

Success criteria

I can:

- ☐ read (x, y) pairs from a table
- ☐ plot the points and join them with a ruler
- ☐ describe the pattern (straight line, vertical, horizontal, increasing, decreasing)

Key terms Write the definition as your teacher explains it.

Table of values

Pattern

Notes

Level 1 · Getting started

Easy · 12 questions

WE DO

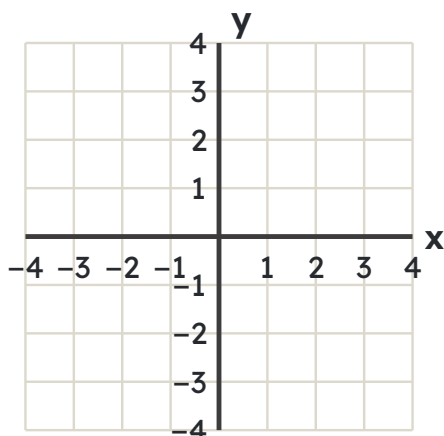
With your teacher: copy what they write on the board.

Example 1

Plot the table, then describe the pattern.

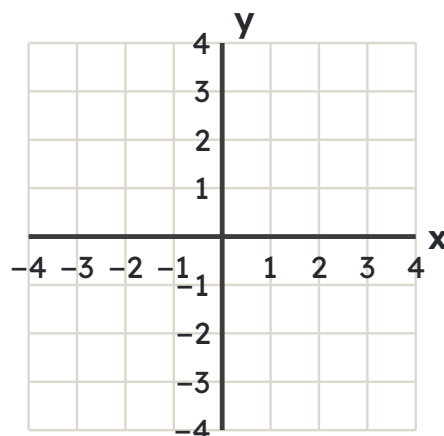
a Plot the table and describe the pattern.

x	-2	-2	-2	-2
y	-2	-1	0	1



b Plot and describe.

x	-2	-1	0	2
y	3	3	3	3



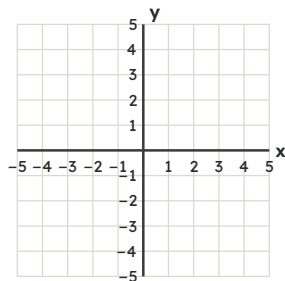
YOU DO On your own: **Easy** Set A. Plot each table and join the points.

2 QUESTIONS

REMEMBER Each column of a table is a point (x, y).

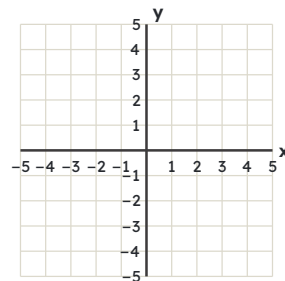
1

x	-2	-2	-2	-2
y	-2	-1	0	1



2

x	-2	-1	0	2
y	3	3	3	3



YOU DO On your own: **Easy** Set A2. Quick drill: write each table column as a point (x, y).

6 QUESTIONS

- 1 $x = 0, y = 3$ 2 $x = 2, y = -1$ 3 $x = -4, y = 0$
- 4 $x = -1, y = -5$ 5 $x = 5, y = 5$ 6 $x = 0, y = 0$

YOU DO On your own: **Easy** Set B. Write the table as a list of points.

4 QUESTIONS

1

x	0	1	2
y	5	4	3

3

x	1	2	3
y	2	4	6

2

x	-1	0	1
y	-1	1	3

4

x	0	2	4
y	1	1	1

SKILL DRILL 1 Easy basics: practise until it is automatic

25 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____

Finish time _____

Score _____ / 25

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–10: write each column of the table as a point (x, y).

1 x: 0, 1, 2, 3
y: 5, 9, 13, 17
_____2 x: 0, 1, 2, 3
y: 0, 3, 6, 9
_____3 x: 0, 1, 2, 3
y: 5, 7, 9, 11
_____4 x: 0, 1, 2, 3
y: 2, 3, 4, 5
_____5 x: 0, 1, 2, 3
y: 1, 3, 5, 7
_____6 x: 0, 1, 2, 3
y: 3, 7, 11, 15
_____7 x: 0, 1, 2, 3
y: 6, 10, 14, 18
_____8 x: 0, 1, 2, 3
y: 3, 6, 9, 12
_____9 x: 0, 1, 2, 3
y: 1, 4, 7, 10
_____10 x: 0, 1, 2, 3
y: 5, 6, 7, 8

ROUND 2 Questions 11–25: the points follow a pattern. Write the next point.

11 (3, 13), (4, 17), (5, 21)
_____12 (1, 7), (2, 10), (3, 13)
_____13 (1, 1), (2, 2), (3, 3)
_____14 (3, 18), (4, 22), (5, 26)
_____15 (0, 4), (1, 8), (2, 12)
_____16 (0, 2), (1, 3), (2, 4)
_____17 (0, 6), (1, 9), (2, 12)
_____18 (1, 5), (2, 9), (3, 13)
_____19 (3, 9), (4, 11), (5, 13)
_____20 (1, 3), (2, 6), (3, 9)
_____21 (0, 5), (1, 8), (2, 11)
_____22 (3, 6), (4, 7), (5, 8)
_____23 (0, 4), (1, 6), (2, 8)
_____24 (1, 6), (2, 9), (3, 12)
_____25 (3, 16), (4, 20), (5, 24)

Level 2 · Building up

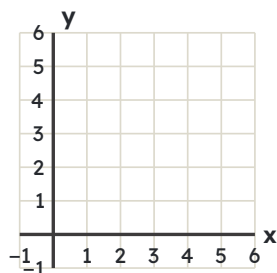
Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Plot the table, then describe the pattern.

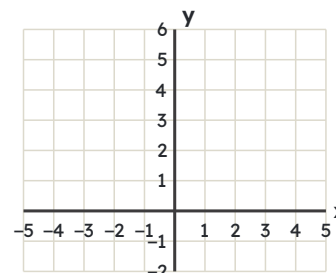
a Plot and describe.

x	0	1	2	5
y	5	4	3	0



b Plot and describe.

x	-1	0	1	2
y	-1	1	3	5



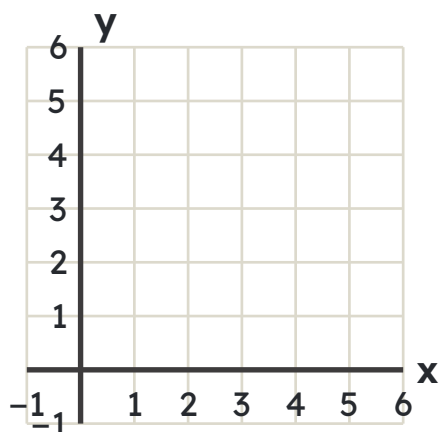
YOU DO On your own: **Medium** Set C. Plot, join and describe the pattern.

2 QUESTIONS

REMEMBER Plot each point carefully.

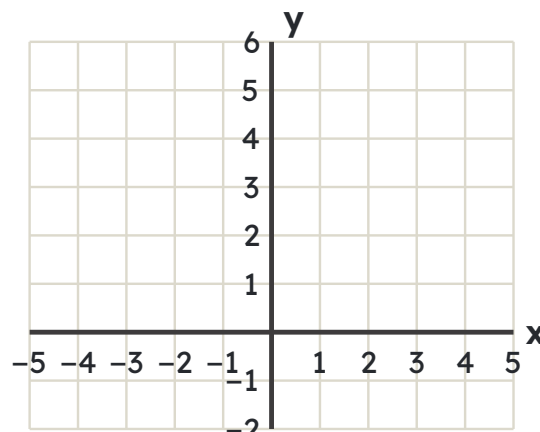
1

x	0	1	2	5
y	5	4	3	0



2

x	-1	0	1	2
y	-1	1	3	5



SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: describe the pattern: what happens to y each time x goes up by 1?

1 (0, 9), (1, 8), (2, 7)

2 (0, 1), (1, 4), (2, 7)

3 (0, 5), (1, 7), (2, 9)

4 (0, 5), (1, 3), (2, 1)

5 (0, 9), (1, 6), (2, 3)

6 (0, 4), (1, 6), (2, 8)

7 (0, 9), (1, 12), (2, 15)

8 (0, 1), (1, 2), (2, 3)

9 (0, 0), (1, 4), (2, 8)

10 (0, 5), (1, 10), (2, 15)

11 (0, 2), (1, -1), (2, -4)

12 (0, 8), (1, 12), (2, 16)

ROUND 2 Questions 13–27: the points follow a pattern. Find the missing coordinate.

13 (0, -1), (1, 4), (2, 9), (5, ?)

14 (0, -2), (1, 1), (2, 4), (5, ?)

15 (0, 0), (1, 3), (2, 6), (5, ?)

16 (0, 4), (1, 7), (2, 10), (5, ?)

17 (0, 4), (1, 9), (2, 14), (5, ?)

18 (0, 3), (1, 8), (2, 13), (5, ?)

19 (0, -2), (1, -1), (2, 0), (5, ?)

20 (0, 4), (1, 8), (2, 12), (5, ?)

21 (0, 0), (1, 5), (2, 10), (5, ?)

22 (0, 0), (1, 2), (2, 4), (5, ?)

23 (0, -1), (1, 1), (2, 3), (5, ?)

24 (0, -1), (1, 3), (2, 7), (5, ?)

25 (0, 3), (1, 4), (2, 5), (5, ?)

26 (0, 3), (1, 5), (2, 7), (5, ?)

27 (0, -2), (1, 0), (2, 2), (5, ?)

Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Find the pattern, then predict.

a In the table in Example 4, what happens to y when x goes up by 1?

Your teacher works this out. Copy every step.

b Predict y when $x = 3$ in Example 4.

Your teacher works this out. Copy every step.

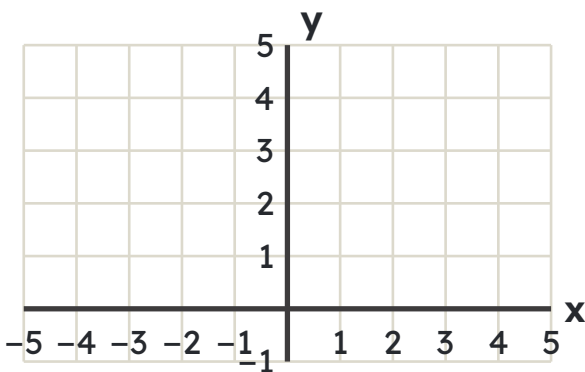
YOU DO On your own: **Challenging** Set D. Plot and describe.

2 QUESTIONS

REMEMBER Join them with a ruler if they line up.

1

x	-2	-1	0	1	2
y	4	1	0	1	4



2 Is the pattern in Question 1 a straight line? Describe it.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

The points (0, 2), (1, 5), (2, 8) follow a pattern.
Find y when $x = 10$.

1. y goes up by 3 each time x goes up by 1
2. Start at 2 when $x = 0$: $y = 3x + 2$
3. $3 \times 10 + 2$

Answer: 32

YOU DO

On your own: read the worked example, then try these.

- E1 The points (0, 7), (1, 5), (2, 3) follow a pattern.
Find y when $x = 6$.

- E2 Find the rule for the points (1, 4), (2, 8), (3, 12).

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Write the points in this table: $x = 0, 1, 2$ and $y = 3, 5, 7$.

Level 2

Describe the pattern of the points (1, 4), (2, 4), (3, 4).

Level 3

What happens to y each time x goes up by 1 for (0, 1), (1, 4), (2, 7)?

Lesson summary

Missed the lesson? Start here.

Remember

1. Each column of a table is a point (x, y).
2. Plot each point carefully.
3. Join them with a ruler if they line up.
4. Describe: horizontal, vertical, going up or going down.

Level 1

x always -2

1. All points above/below each other

Answer: vertical line

Level 2

y always 3

1. All points level

Answer: horizontal line

Level 3

(0, 5), (1, 4), (2, 3)

1. y goes down 1

Answer: line going down

Extension

(0, 2), (1, 5), (2, 8)

1. $y = 3x + 2$

Answer: 32 when $x = 10$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.05

Tables of values from rules

Date _____

Learning intentions

We are learning to:

- complete a table of values from a rule

Success criteria

I can:

- ☐ substitute each x value into the rule
- ☐ use negative numbers carefully
- ☐ write the rule from a table

Key terms Write the definition as your teacher explains it.

Rule

Substitute

Notes

Level 1 · Getting started

Easy · 15 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Complete each table from its rule.

a Complete for $y = x + 3$.

x	-1	0	1	2	3
y					

b Complete for $q = 2p - 7$.

p	-2	-1	0	1	2
q					

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Complete each table.

4 QUESTIONS

REMEMBER Write the rule.

① $y = x + 3$

x	-1	0	1	2	3
y					

③ $y = x - 4$

x	0	1	2	3	4
y					

② $y = 2x$

x	-1	0	1	2	3
y					

④ $y = 5x$

x	-2	-1	0	1	2
y					

YOU DO On your own: **Easy** Set A2. Quick drill: find y.

9 QUESTIONS

① $y = x + 4, x = 2$

② $y = 3x, x = 5$

③ $y = x - 7, x = 1$

④ $y = 2x + 1, x = 3$

⑤ $y = 10 - x, x = 4$

⑥ $y = 4x - 2, x = 0$

⑦ $y = -x, x = -3$

⑧ $y = 2x - 5, x = -1$

⑨ $y = x \div 2, x = 8$

YOU DO On your own: **Easy** Set B. Complete each table.

2 QUESTIONS

① $q = 2p - 7$

p	-2	-1	0	1	2
q					

② $d = 5 - c$

c	-1	0	1	2	3
d					

SKILL DRILL 1 Easy basics: practise until it is automatic

34 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 34

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: substitute to find y .

1 $y = 2x + 1, x = 7$

2 $y = 4x + 6, x = 2$

3 $y = 4x + 8, x = 2$

4 $y = 5x + 1, x = 1$

5 $y = 6x - 2, x = 4$

6 $y = x - 2, x = 4$

7 $y = 5x, x = 1$

8 $y = 3x - 4, x = 3$

9 $y = 4x + 3, x = 4$

10 $y = 4x - 5, x = 5$

11 $y = 5x - 1, x = 0$

12 $y = 5x, x = 3$

13 $y = 2x + 3, x = 1$

14 $y = 6x + 3, x = 3$

15 $y = 2x + 2, x = 6$

16 $y = 5x + 5, x = 8$

17 $y = x + 6, x = 4$

18 $y = 2x + 2, x = 8$

19 $y = 4x + 9, x = 7$

20 $y = 6x - 2, x = 1$

21 $y = 5x, x = 4$

22 $y = 6x + 2, x = 4$

23 $y = x, x = 4$

24 $y = x + 5, x = 7$

ROUND 2 Questions 25–34: complete the table: write the y -values for $x = 0, 1, 2, 3$.

25 $y = 3x + 8$

26 $y = 4x + 4$

27 $y = x + 8$

28 $y = 3x + 9$

29 $y = 4x + 3$

30 $y = x + 6$

31 $y = 4x + 6$

32 $y = 5x$

33 $y = 2x + 3$

34 $y = x + 9$

Level 2 · Building up

Medium · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Complete each table from its rule.

a Complete for $d = 5 - c$.

c	-1	0	1	2	3
d					

Your teacher works this out. Copy every step.

b Complete for $y = 3x + 1$.

x	-2	-1	0	1	2
y					

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Complete each table.

4 QUESTIONS

REMEMBER Replace x with each value (use brackets for negatives).

1 $y = 3x + 1$

x	-2	-1	0	1	2
y					

3 $y = \frac{1}{2}x - 2$

x	-4	-2	0	2	4
y					

2 $y = -2x + 3$

x	-2	-1	0	1	2
y					

4 $y = 10 - 3x$

x	0	1	2	3	4
y					

SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: find the rule, $y = \dots x + \dots$

1 $x: 0, 1, 2, 3$
 $y: 1, 2, 3, 4$

2 $x: 0, 1, 2, 3$
 $y: 6, 11, 16, 21$

3 $x: 0, 1, 2, 3$
 $y: 2, 3, 4, 5$

4 $x: 0, 1, 2, 3$
 $y: 0, 4, 8, 12$

5 $x: 0, 1, 2, 3$
 $y: 4, 5, 6, 7$

6 $x: 0, 1, 2, 3$
 $y: 4, 6, 8, 10$

7 $x: 0, 1, 2, 3$
 $y: 0, 6, 12, 18$

8 $x: 0, 1, 2, 3$
 $y: 0, 3, 6, 9$

9 $x: 0, 1, 2, 3$
 $y: 2, 4, 6, 8$

10 $x: 0, 1, 2, 3$
 $y: 7, 8, 9, 10$

11 $x: 0, 1, 2, 3$
 $y: 6, 12, 18, 24$

12 $x: 0, 1, 2, 3$
 $y: 1, 5, 9, 13$

ROUND 2 Questions 13–27: find x for the value of y given.

13 $y = 5x + 5, y = 15$

14 $y = 5x + 3, y = 48$

15 $y = 3x + 5, y = 29$

16 $y = 4x + 7, y = 11$

17 $y = 4x + 8, y = 28$

18 $y = 4x + 1, y = 33$

19 $y = 2x + 2, y = 10$

20 $y = 3x + 4, y = 7$

21 $y = 6x + 8, y = 8$

22 $y = 6x + 3, y = 39$

23 $y = 5x, y = 30$

24 $y = 3x + 7, y = 28$

25 $y = 3x + 4, y = 34$

26 $y = 4x + 7, y = 7$

27 $y = 4x + 2, y = 38$

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Find the rule for each table.

a Find the rule.

x	0	1	2	3
y	4	6	8	10

Your teacher works this out. Copy every step.

b Find the rule.

x	1	2	3	4
y	7	6	5	4

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Find the rule for each table.

4 QUESTIONS

REMEMBER Calculate y.

1

x	0	1	2	3
y	4	6	8	10

3

x	0	1	2	3
y	-3	1	5	9

2

x	1	2	3	4
y	7	6	5	4

4

x	-1	0	1	2
y	3	0	-3	-6



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

Complete a table for $y = x^2 - 1$ using $x = -2$ to 2.

1. Square first, then subtract 1
2. $(-2)^2 - 1 = 3$

Answer: $y = 3, 0, -1, 0, 3$

YOU DO

On your own: read the worked example, then try these.

- E1 Complete a table for $y = x^2 + x$ using $x = -2$ to 2.

- E2 Find the rule: $x = 1, 2, 3, 4$ and $y = 1, 4, 9, 16$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

For $y = x + 5$, find y when $x = -2$.

Level 2

For $y = 4x - 1$, complete for $x = 0, 1, 2$.

Level 3

Find the rule: $x = 0, 1, 2$ and $y = 1, 4, 7$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Write the rule.
2. Replace x with each value (use brackets for negatives).
3. Calculate y .
4. To find a rule: how much does y change when x goes up 1? Then find the start value.

Level 1

$$y = x + 3$$

1. $-1 + 3 = 2, \dots$

Answer: 2, 3, 4, 5, 6

Level 2

$$q = 2p - 7$$

1. $2 \times (-2) - 7 = -11$

Answer: -11, -9, -7, -5, -3

Level 3

Rule for 4, 6, 8, 10

1. +2 each time, start 4

Answer: $y = 2x + 4$

Extension

$$y = x^2 - 1$$

1. Square first

Answer: 3, 0, -1, 0, 3

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.06

Graphing linear equations

Date _____

Learning intentions

We are learning to:

- graph a linear equation using a table of values
- test whether a point lies on a line

Success criteria

I can:

- ☐ complete a table of values for the equation
- ☐ plot the points and rule a straight line through them
- ☐ substitute a point into the equation to test it

Key terms Write the definition as your teacher explains it.

Linear

Equation

Notes

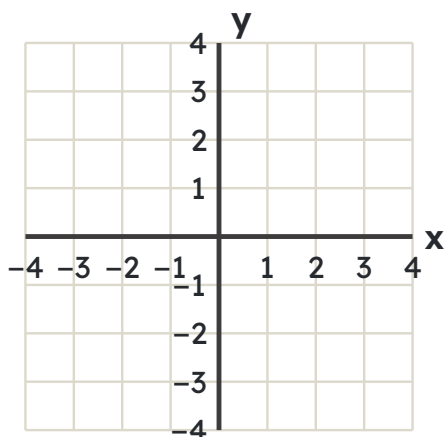
Level 1 · Getting started

Easy · 14 questions

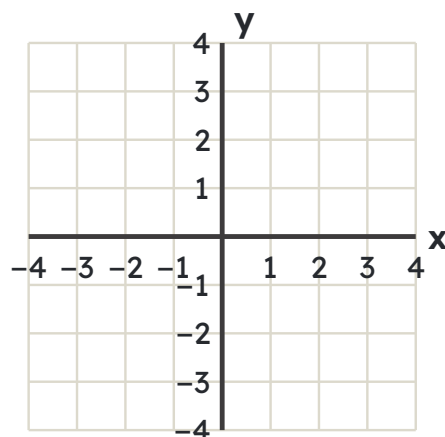
WE DO With your teacher: copy what they write on the board.

Example 1 Complete a table, then graph the line.

a Graph $y = 2x - 1$ for $x = -1$ to 2.



b Graph $y = -x + 3$.

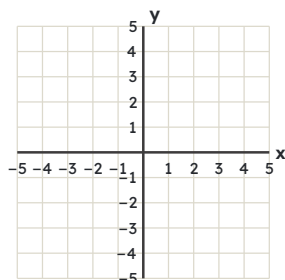


YOU DO On your own: **Easy** Set A. Complete a table, then graph.

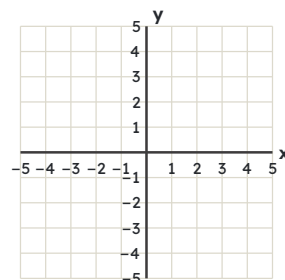
2 QUESTIONS

REMEMBER Make a table with 3 or more x values.

1 $y = x + 1$ ($x = -2$ to 2)



2 $y = 2x - 1$ ($x = -1$ to 2)



YOU DO On your own: **Easy** Set A2. Quick drill: does the point lie on $y = 2x + 1$? Write yes or no.

6 QUESTIONS

1 $(0, 1)$

2 $(1, 3)$

3 $(2, 4)$

4 $(-1, -1)$

5 $(3, 7)$

6 $(-2, -4)$

YOU DO On your own: **Easy** Set B. Does the point lie on $y = 3x + 2$? Write yes or no.

6 QUESTIONS

1 $(1, 2)$

4 $(0, 2)$

2 $(-1, 4)$

5 $(1, 5)$

3 $(-2, -4)$

6 $(2, 8)$

SKILL DRILL 1 Easy basics: practise until it is automatic

32 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 32 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: does the point lie on the line? Write yes or no.

1 (0, 2) on $y = 4x + 2$

2 (2, 7) on $y = 2x + 3$

3 (3, 13) on $y = 4x + 1$

4 (–3, 2) on $y = x + 3$

5 (3, 14) on $y = 3x + 5$

6 (0, 1) on $y = x + 1$

7 (–1, 1) on $y = 3x + 4$

8 (3, 8) on $y = x + 4$

9 (0, –1) on $y = 4x – 1$

10 (2, 10) on $y = 3x + 2$

11 (0, 4) on $y = 3x + 2$

12 (–3, –7) on $y = 3x + 2$

13 (2, 5) on $y = x + 1$

14 (1, 5) on $y = 2x + 3$

15 (4, 13) on $y = 2x + 5$

16 (–1, 2) on $y = 2x + 2$

17 (5, 9) on $y = x + 4$

18 (2, 11) on $y = 4x + 3$

19 (0, 6) on $y = 3x + 5$

20 (1, 7) on $y = 2x + 5$

21 (–3, –2) on $y = x + 1$

22 (–3, 0) on $y = 2x + 4$

23 (–3, –5) on $y = 2x – 2$

24 (–1, 1) on $y = 2x + 3$

ROUND 2 Questions 25–32: write the points to plot for $x = -1, 0, 1, 2$.

25 $y = 2x – 1$

26 $y = 3x – 1$

27 $y = x + 3$

28 $y = 2x + 2$

29 $y = 3x + 1$

30 $y = x$

31 $y = 3x$

32 $y = 3x + 2$

Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Does the point lie on the line?

a Does $(1, 5)$ lie on $y = 3x + 2$?

Your teacher works this out. Copy every step.

b Does $(-2, -4)$ lie on $y = 3x + 2$?

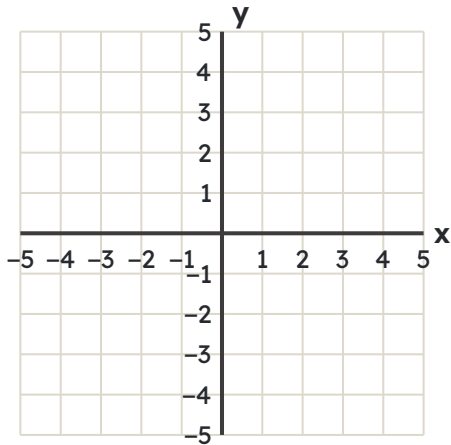
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Complete a table, then graph.

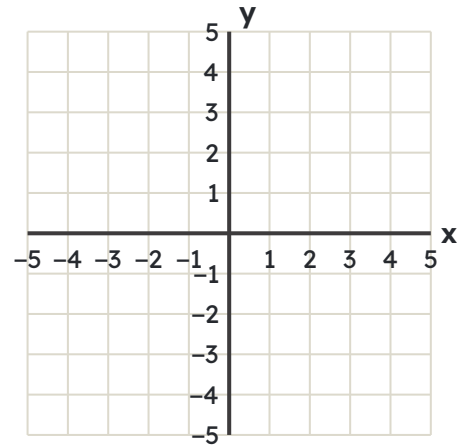
2 QUESTIONS

REMEMBER Substitute to find y .

1 $y = -x + 3$



2 $y = \frac{1}{2}x - 2$



SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: the point lies on the line. Find the missing coordinate.

1 (0, ?) on $y = -2x - 4$

2 (-1, ?) on $y = x + 4$

3 (0, ?) on $y = 2x$

4 (2, ?) on $y = 2x$

5 (4, ?) on $y = -x - 2$

6 (-2, ?) on $y = 2x + 4$

7 (3, ?) on $y = 4x - 1$

8 (0, ?) on $y = 2x + 2$

9 (-1, ?) on $y = 4x$

10 (-3, ?) on $y = 2x - 4$

11 (4, ?) on $y = -x + 3$

12 (2, ?) on $y = 3x + 6$

13 (3, ?) on $y = 4x - 3$

14 (4, ?) on $y = 4x + 3$

15 (0, ?) on $y = 3x - 1$

16 (-4, ?) on $y = -3x - 2$

ROUND 2 Questions 17–28: where does the line cut the y-axis? Write the point.

17 $y = -4x - 3$

18 $y = -4x - 1$

19 $y = -2x - 2$

20 $y = 3x - 5$

21 $y = 3x + 7$

22 $y = -2x + 6$

23 $y = 2x + 9$

24 $y = 4x + 3$

25 $y = -3x + 9$

26 $y = -2x - 6$

27 $y = 2x$

28 $y = -3x - 3$

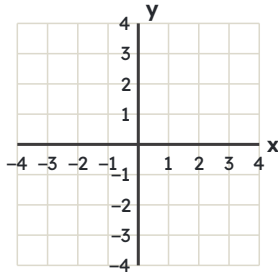
Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Graph the line, then answer.

a Graph $y = \frac{1}{2}x - 2$ for $x = -4$ to 4 .



b Where does $y = 2x - 1$ cross the y-axis?

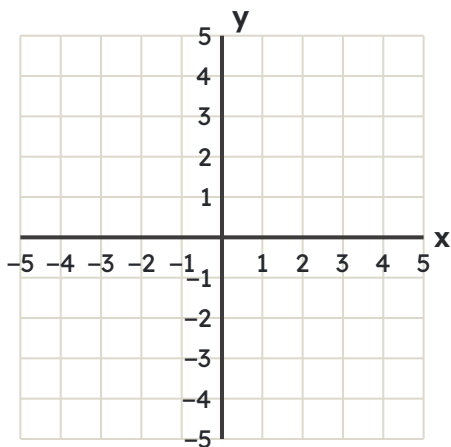
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Graph and compare.

2 QUESTIONS

REMEMBER Plot and rule a line through all the points.

1 Graph $y = x$ and $y = x + 2$ on the same plane. What do you notice?



2 Where do $y = x + 1$ and $y = -x + 3$ meet? (Use your graphs.)

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked exampleFind the value of k if $(2, k)$ lies on $y = 4x - 3$.

1. Substitute $x = 2$
2. $k = 4 \times 2 - 3$

Answer: $k = 5$ **YOU DO**

On your own: read the worked example, then try these.

- E1** Find m if $(m, 11)$ lies on $y = 2x + 5$.

- E2** Which of these lines are parallel: $y = 2x + 1$, $y = x + 2$, $y = 2x - 3$?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO**Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1For $y = x + 4$, find y when $x = 1$.**Level 2**Does $(2, 5)$ lie on $y = 2x + 1$?**Level 3**Does $(-1, 3)$ lie on $y = -2x + 1$?

Lesson summary

Missed the lesson? Start here.

Remember

1. Make a table with 3 or more x values.
2. Substitute to find y .
3. Plot and rule a line through all the points.
4. Test a point: substitute x and see if you get its y .

Level 1

$$y = 2x - 1$$

1. $x: -1, 0, 1, 2 \rightarrow y: -3, -1, 1, 3$

Answer: a straight line through $(0, -1)$

Level 2

$$(1, 5) \text{ on } y = 3x + 2?$$

1. $3 \times 1 + 2 = 5 \checkmark$

Answer: yes

Level 3

$$(-2, -4)$$

1. $3 \times (-2) + 2 = -4 \checkmark$

Answer: yes

Extension

$$(2, k) \text{ on } y = 4x - 3$$

1. $8 - 3$

Answer: $k = 5$

My reflection: what did I learn today? Which question was hardest, and why?

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